

Frontend – Assignment 1

Q1. HTTP Request–Response Cycle

The HTTP Request–Response Cycle is the process through which a browser communicates with a web server to get a web resource such as an HTML page, CSS file, image, or data. The main steps are:

- 1. User requests a resource:** The user enters a URL or clicks a link in the browser.
- 2. Browser sends a request:** The browser creates an HTTP request containing information such as the requested URL and HTTP method.
- 3. Server receives the request:** The web server processes the request and finds the required resource or performs the requested operation.
- 4. Server sends a response:** The server returns an HTTP response containing a status code, headers, and usually the requested content.
- 5. Browser receives and displays the response:** The browser processes the response and renders the webpage for the user.

Client / Browser	Web Server
1. HTTP Request →	Receives and processes request
← 2. HTTP Response	Sends status + content
Browser renders webpage	

Q2. Introduction to Web Technologies

HTML: HTML (HyperText Markup Language) is used to create the structure of a webpage. It defines elements such as headings, paragraphs, links, images, tables, and forms.

CSS: CSS (Cascading Style Sheets) is used to style and design webpages. It controls colors, fonts, spacing, borders, layouts, and responsiveness.

JavaScript: JavaScript is a programming language used to add interactivity and dynamic behavior to webpages. It can respond to user actions, change content, validate forms, and update the page.

Bootstrap: Bootstrap is a frontend framework that provides ready-made CSS classes and components for creating responsive and attractive webpages more quickly.

jQuery: jQuery is a JavaScript library that simplifies common tasks such as DOM manipulation, event handling, animations, and AJAX requests.

React.js: React.js is a JavaScript library used for building user interfaces. It helps developers create reusable components and efficiently update parts of a webpage.

Q3. Client-Side vs. Server-Side Technologies

Client-side technologies run mainly in the user's browser. They are responsible for what the user sees and interacts with. Examples include HTML, CSS, and JavaScript.

Server-side technologies run on a web server. They process requests, perform application logic, work with databases, and send a response back to the client. Examples include Java, Python, PHP, Node.js,

and .NET.

Basis	Client-Side	Server-Side
Runs on	User's browser	Web server
Main purpose	Interface and interaction	Business logic and data processing
Examples	HTML, CSS, JavaScript	Java, Python, PHP, Node.js
Visible to user	Usually affects the visible webpage	Usually works behind the scenes

Q4. Block-Level Elements in HTML

Block-level elements are HTML elements that normally start on a new line and take up the available width of their parent container. They are useful for arranging the main sections of a webpage. Examples include `<div>`, `<p>`, and `<h1>`.

Characteristics: They generally begin on a new line, occupy the available horizontal space, and can contain other elements depending on the element's rules.

Q5. Semantic HTML Elements

Semantic HTML elements clearly describe the meaning or purpose of their content. They make the structure of a webpage easier to understand for developers, browsers, and assistive technologies. They can also improve accessibility and make code easier to maintain.

Examples: `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<aside>`, and `<footer>`.

Q6. Creating a Table Using HTML

```
<!DOCTYPE html>
<html>
<head>
<title>Employee Table</title>
</head>
<body>

<table border="1">
<tr>
<th>Sr. No.</th>
<th>Name</th>
<th>Designation</th>
<th>Department</th>
</tr>
<tr>
<td>1</td>
<td>John</td>
<td>Manager</td>
<td>Sales</td>
</tr>
<tr>
<td>2</td>
<td>Smith</td>
<td>Sr. Manager</td>
<td>Marketing</td>
</tr>
<tr>
<td>3</td>
<td>Jane</td>
<td>Manager</td>
<td>HR</td>
</tr>
```

```
</table>

</body>
</html>
```

Q7. Creating a Nested List Using HTML

```
<!DOCTYPE html>
<html>
<body>

<ul>
<li>River
<ul>
<li>Ganga</li>
<li>Yamuna</li>
<li>Brahmaputra</li>
<li>Narmada</li>
</ul>
</li>
</ul>

</body>
</html>
```

Q8. Types of CSS

1. Inline CSS: CSS is written directly inside the HTML element using the style attribute.

Example: `<p style="color: blue;">Hello</p>`

2. Internal CSS: CSS is written inside a `<style>` element in the `<head>` section of the HTML document.

Example: `<style> p { color: blue; } </style>`

3. External CSS: CSS is written in a separate .css file and connected to HTML using a `<link>` element.

Example: `<link rel="stylesheet" href="style.css">`

Q9. Practical Understanding

Technology	Role in a Web Page
HTML	Provides the basic structure and content of the webpage.
CSS	Controls the appearance, layout, spacing, colors, fonts, and overall presentation.
JavaScript	Adds logic, interaction, and dynamic behavior to the webpage.

In simple terms, HTML creates the structure, CSS makes the page look better, and JavaScript makes the page interactive.

Q10. Short Answer

a. What is the role of Bootstrap in web development?

Bootstrap provides ready-to-use CSS classes and components that help developers create responsive and consistent webpages quickly. It is especially useful for layouts, buttons, forms, navigation bars, and other common interface elements.

b. What is jQuery and why was it widely used in web development?

jQuery is a JavaScript library that makes common JavaScript tasks simpler. It was widely used because it made DOM manipulation, event handling, animations, and AJAX operations easier with less code and

helped handle browser differences.

c. What is React.js and what problem does it solve in frontend development?

React.js is a JavaScript library for building user interfaces using reusable components. It helps developers manage and update complex interfaces efficiently without having to manually change every part of the page.